higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T180**(E)**(A3)T

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL SURVEYING N6

(8060056)

3 April 2019 (X-Paper)
09:00–12:00

Calculators may be used.

This question paper consists of 4 pages, 2 addenda, 1 diagram sheet and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING AND STRUCTURAL SURVEYING N6
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Sketches should be neatly and clearly labelled.
6. Write neatly and legibly.

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QUESTION 1

Calculate the coordinates of R1, R2 and R3 on ADDENDUM A (attached) from the following data (adjustments must be made according to the Bowditch rule):

Coordinates:

P	+458,82	+1 532,32
Q	-38,78	+1 733,58

Line	Direction	Distance (m)
P-R1	232:39:10	248,04
R1-R2	286:56:08	270,21
R2-R3	327:11:36	213,63
R3-Q	38:41:50	119,17

[20]

QUESTION 2

The notes below refer to observations from T in a tacheometric survey. The elevation of the survey station T is 713,24 m and the theodolite is 1,46 m above T. The booked vertical angles are zenith distance.

Staff station	Horizontal angle	Vertical angle	Stadia readings
T1	136:49:00	98:36:00	1,90…….0,58
T2	173:21:00	81:39:00	4,02…….2,60
T3	246:32:00	82:09:00	1,84…….0,60
T4	299:52:00	101:36:00	3,64…….1,26

Use the given information to complete the tacheometric sheet on ADDENDUM B (attached). **[20]**

QUESTION 3

Explain each of the following surveying terms:

3.1 Survey station

3.2 Level line

3.3 Contours

3.4 Booking

3.5 Staking

(5 × 2) **[10]**

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QUESTION 4

The following information applies to FIGURE 1 on the DIAGRAM SHEET (attached):

Coordinates	Distances	Angles
A +4 896,26 +3 926,82	A-C 343,55 m	BAC 26:25:31
B +4 328,66 +3 844,24	B-C 306,74 m	ABC 29:53:47

4.1 Calculate the direction of AB, BC and AC. (6)

4.2 Calculate and check the coordinates of C. (8)

[14]

QUESTION 5

FIGURE 2 on the DIAGRAM SHEET (attached) represents the plan of an area to be excavated to a level of 125,00 m.

Calculate the volume of ground to be carted away using a bulking factor of 15%. **[10]**

QUESTION 6

Calculate from the data given below the setting out data for a road curve:

- A peg is required at every full 20 m chainage.
- The curve is to the left.
- The radius is 218,50 m.
- The angle of intersection (Δ) is 39:21:00.
- Changing at the intersection point is 3 372,33 m. **[20]**

QUESTION 7

From the readings below taken by a theodolite setup at station A, deduce the correct angles of elevation to B2 and B5. Clearly show whether the angles are rise or fall.

At A:		
Station	**Circle left**	**Circle right**
B2	129:44:50	230:13:48
B5	81:08:24	278:52:18

[6]

TOTAL: 100

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ADDENDUM A **EXAMINATION NUMBER:**

NAME	JOIN	ΔY	ΔX	NAME	Y	X
P				P	+458,82	+1 532,32
232:39:10						
248,04 m						
R1				R1		
286:56:08						
270,21 m						
R2				R2		
327:11:36						
213,63 m						
R3				R3		
38:41:50						
119,17 m						
Q				Q	-38,78	+1 733,58

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ADDENDUM B **EXAMINATION NUMBER:** | | | | | | | | | | | | | |

Station		Distance		HI or middle hair MH	Angles		HI – MH + -	Height component + -	Height difference + -	Elevation of point	Remarks
From	To	Stadia	Hor		Hor	Vert					
T				1,46						713,24	
	T1	1,90 0,58			136:49:00	98:36:00					
	T2	4,02 2,60			173:21:00	81:39:00					
	T3	1,84 0,60			246:32:00	82:09:00					
	T4	3,64 1,26			299:52:00	101:36:00					

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DIAGRAM SHEET

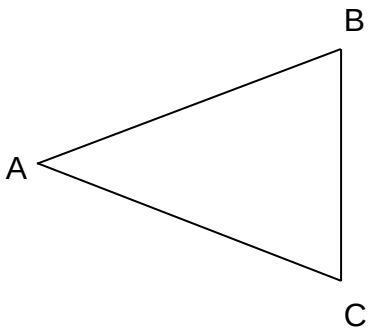


FIGURE 1

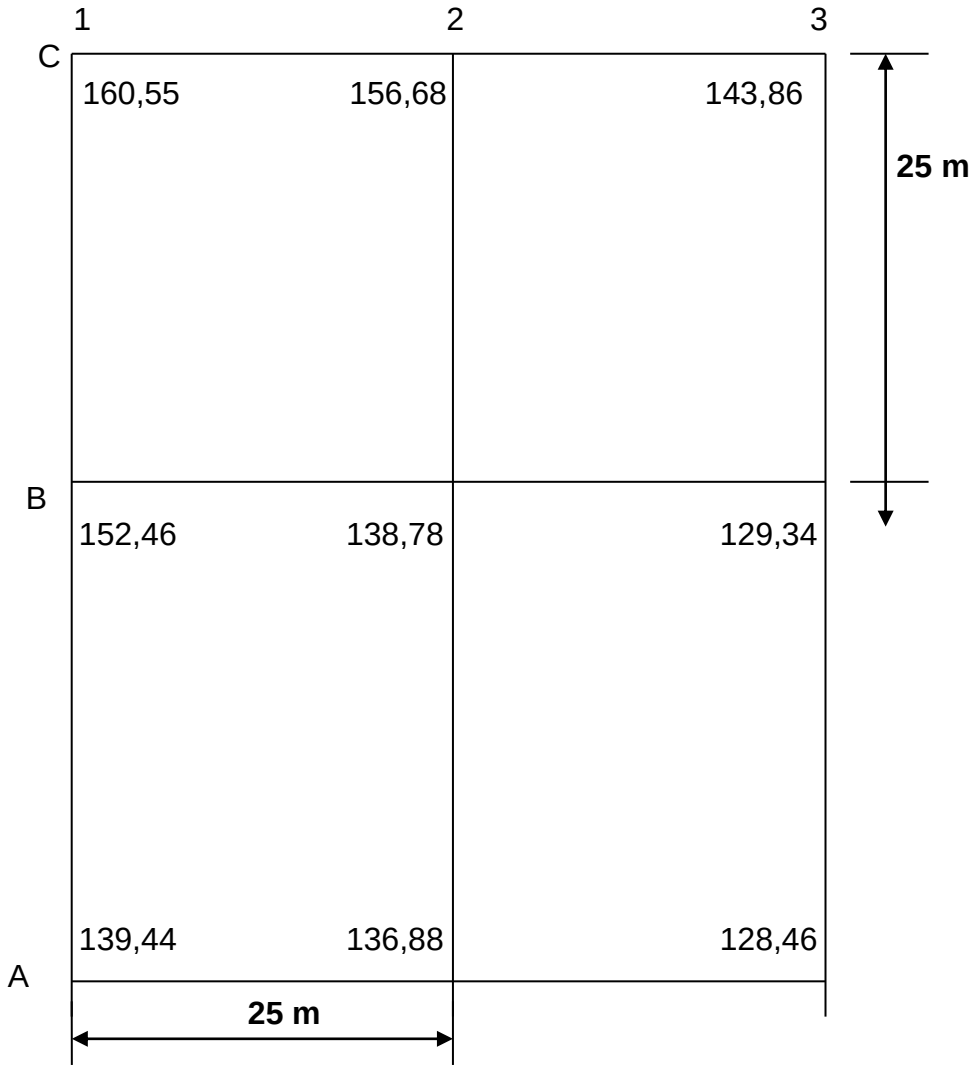


FIGURE 2

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FORMULA SHEET

Any other applicable formula may also be used.

$$\alpha = \tan^{-1}\frac{\Delta y}{\Delta x}$$

$$\alpha = \tan^{-1}\frac{\Delta x}{\Delta y} + 90^{\circ}$$

$$\alpha = \tan^{-1}\frac{\Delta y}{\Delta x} + 180^{\circ}$$

$$\alpha = \tan^{-1}\frac{\Delta x}{\Delta y} + 270^{\circ}$$

$$S = \frac{\Delta y}{\sin\alpha}$$

$$S = \frac{\Delta x}{\cos\alpha}$$

$$\Delta y = s.\sin\alpha$$

$$\Delta x = s.\cos\alpha$$

$$C = \frac{Distance}{Total\ \ distance} X_l$$

$$\Delta h = 50I\sin 2\theta + HI - MH = 100I\sin\theta\cos\theta + HI - MH$$

$$HD = 100I\cos^2\theta$$

$$T = R.\tan\frac{\Delta}{2}$$

$$La = \frac{\pi.\Delta.R}{180}$$

$$\vartheta = \frac{1718{,}9.a}{R}$$

$$Cd = T.\tan\frac{\Delta}{4}$$

$$Lc = 2.R.\sin\frac{\Delta}{2}$$

$$W_1 = \frac{g(a+hs)}{g-s}$$

$$W_2 = \frac{g(a+hs)}{g+s}$$

$$A = \frac{W_1.W_2 - a^2}{s}$$

$$a^2 = b^2 + c^2 - 2bc.CosA$$

$$b^2 = a^2 + c^2 - 2ac.CosB$$

$$c^2 = a^2 + b^2 - 2ab.CosC$$

$$\frac{SinA}{a} = \frac{SinB}{b} = \frac{SinC}{c}$$

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